

# KANGYOON LEE

klee320@jhu.edu | Homepage

## RESEARCH INTERESTS

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### ***Causal Network Traffic Modeling for Preemptive Congestion Control***

My research aims to enable preemptive congestion control in datacenter networks. I formalize the causal domain knowledge underlying traffic patterns—temporal user behaviors, physical infrastructure states, and structural feature dependencies—and embed these principles as a priori constraints in predictive models. This yields explainable predictions that operators can trust, enabling proactive intervention before congestion impacts users.

## EDUCATION

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| Aug. 2026 – Present   | <b>Johns Hopkins University</b><br>Ph.D. in Computer Science                                                                                                                                                                                                                                                                                                       | Baltimore, MD |
| Mar. 2018 – Feb. 2025 | <b>Hankuk University of Foreign Studies (HUFS)</b><br>B.Eng. in Information and Communications Engineering <ul style="list-style-type: none"><li>GPA: <b>3.96/4.00 (Top 1.8%)</b></li><li>Thesis: <i>A Data Forwarding Scheme Using Proportional-Fair Scheduling Algorithm in Edge Network</i></li><li>Completed 19 months of mandatory military service</li></ul> | Yongin, Korea |

## WORK EXPERIENCE

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| Mar. 2025 – May. 2026 | <b>Research Intern</b><br>Institute: Korea Institute of Science and Technology (Seoul, Republic of Korea)<br>Role: Network traffic analysis and preprocessing; Multi-modal time-series modeling <ul style="list-style-type: none"><li>Built multi-modal network traffic DL pipeline for real-time urban crowd density prediction [3][4]</li></ul> |
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## RESEARCH EXPERIENCE

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| Jun. 2024 – Jun. 2025 | <b>Extracting Structural Causality in Data Feature Dependencies</b><br>Institute: AI Education Institute, Hansung University (Seoul, Republic of Korea) <ul style="list-style-type: none"><li>Embedded Bayesian Network-derived feature causality into CTGAN, reducing causal error by 92% [2]</li></ul>                                            |
| Mar. 2024 – Oct. 2024 | <b>Modeling Physical Infrastructure Causality for Adaptive Traffic Routing</b><br>Institute: Information Systems Lab, HUFS (Yongin, Republic of Korea) <ul style="list-style-type: none"><li>Developed adaptive routing algorithm via traffic-infrastructure causal modeling, reducing latency by 9% [1]</li></ul>                                  |
| Sep. 2023 – Dec. 2023 | <b>Benchmarking Generative Models for Data Synthesis</b><br>Institute: AI Education Institute, HUFS (Yongin, Republic of Korea) <ul style="list-style-type: none"><li>Benchmarked generative models for temporal-causal pattern synthesis in multivariate time-series data [5]</li></ul>                                                            |
| Sep. 2022 – May. 2023 | <b>Uncovering Temporal Causality in Network Traffic Patterns</b><br>Institute: Mobile Distributed Computing Lab, HUFS (Yongin, Republic of Korea) <ul style="list-style-type: none"><li>Validated temporal causality in traffic patterns through LSTM-based prediction, achieving 94% accuracy vs. 3-9% for memoryless statistical models</li></ul> |

## PUBLICATIONS

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[1] E. Kim, **K. Lee**, Y. Kim, H. Jung, H. Choi, I. Kim, and H. Park. Frequency-Aware Multi-Resolution Architecture for High-Resolution Spatiotemporal Map Reconstruction from Sparse Observations. 38<sup>th</sup>

*Workshop on Image Processing and Image Understanding (IPIU 2026)*, 2026, Jeju, Korea (Poster)

[2] Y. Kim, E. Kim, **K. Lee**, H. Park, H. Choi, I. Kim, and H. Jung. Learnable Adjacency Matrix Based Spatio-Temporal Graph Convolutional Network for Traffic Prediction. *Autumn Annual Conference of The Institute of Electronics and Information Engineers (IEIE 2025)*, 2025, Gyeonggi, Korea (Poster)

[3] **K. Lee**, H. Ryou, J. Baek, S. Oh, and I. Doo. A Comparative Study of Synthetic Data with Generative AI Models: Safely Reproducing Sensitive Transportation Worker Data. *National Conference on Advanced Engineering and ICT-Convergence (NCAEIC 2023)*, 2023, Jeju, Korea (Oral - **Best Paper Award**)

## PRESENTATION

**Korea Police World Expo (KPEX) 2025**

Incheon, Republic of Korea

Presenter, Korea Institute of Science and Technology (KIST)

## TEACHING EXPERIENCE

Mar. 2024 – **Network Security Course** at HUFS

Jun. 2024 Instructor: Prof. Seongho Jeong

- Graded assignments and provided feedback for 30+ students with Q&A sessions

Sep. 2021 – **Algorithm Analysis and Design Course** at HUFS

Dec. 2021 Instructor: Prof. Sayed C. Shah

- Provided detailed code reviews and debugging support for 50+ students

## SCHOLARSHIPS

Mar. 2024 President's Scholarship (**Full tuition**), HUFS

Sep. 2023 Hyoryeong Merit Scholarship, Cheong-gwon-sa Scholarship Foundation

Sep. 2023 Academic Merit Scholarship (**One-quarter tuition**), HUFS

Mar. 2023 Dean's Scholarship (**Half tuition**), HUFS

Sep. 2022 President's Scholarship (**Full tuition**), HUFS

## AWARDS

Dec. 2023 Best Paper Award, NCAEIC 2023 – ICT-AES

Dec. 2023 Excellence Award, Capstone Design Project, HUFS

Dec. 2023 Excellence Award, Data Creator Camp, Ministry of Science & ICT

Sep. 2020 Creative Award, National Defense Start-Up Challenge, Top 25/939

## NOTABLE ACTIVITIES

- **Community Volunteer** – Regular Donor, Senior & Youth in Crisis, GFoundation, (Mar. 2025 – Present)
- **Conference Staff**, Korea Internet Conference (KRNet) 2023 – Assisted with program coordination, logistics, and technical support
- **Mentor**, AdvICE (Programming Club), HUFS (Sep. 2022 – Dec. 2022) – Guided students in Python algorithm design and competitive programming
- **Communications Specialist**, Republic of Korea Marine Corps (Jun. 2019 – Jan. 2021)

## TECHNICAL SKILLS

Programming & Development

Python, C, C++, Java, R, Bash, Git, Linux

Networking & Traffic Analysis

Wireshark, hping3, Ostinato, ns-3, TCP/IP

Machine Learning

PyTorch, TensorFlow, Keras, Numpy, Pandas, Matplotlib

Embedded & IoT

Raspberry Pi, Arduino